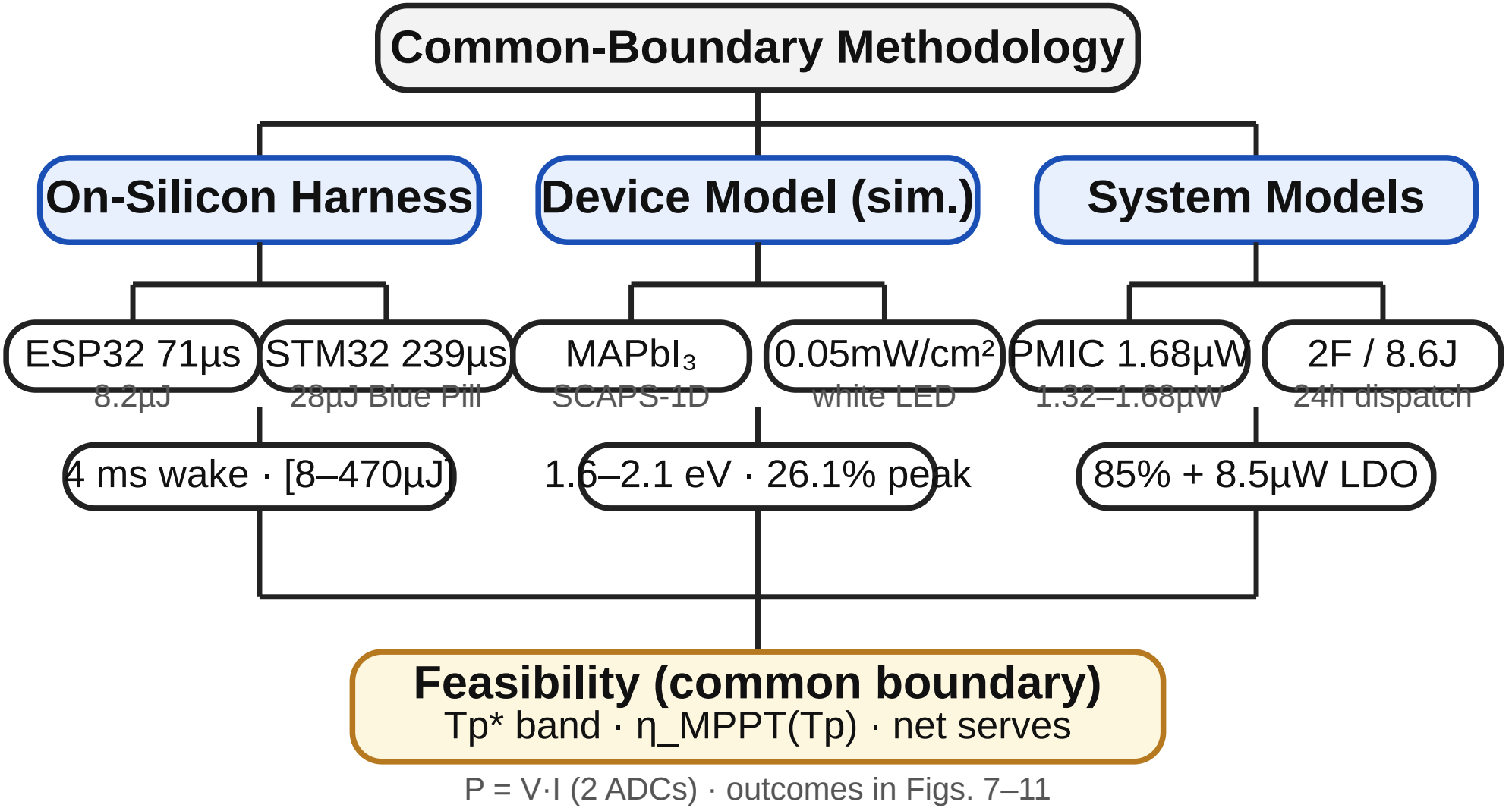
Common-Boundary Methodology
On-Silicon Harness
Device Model (sim.)
System Models
ESP32 71µs
8.2µJ
STM32 239µs
28µJ Blue Pill
MAPbI3
SCAPS-1D
0.05mW/cm²
white LED
PMIC 1.68µW
1.32–1.68µW
2F / 8.6J
24h dispatch
4 ms wake · [8–470µJ]
1.6–2.1 eV · 26.1% peak
85% + 8.5µW LDO
Feasibility (common boundary)
Tp* band · η_MPPT(Tp) · net serves
P = V·I (2 ADCs) · outcomes in Figs. 7–11